

Assignment 02

Deadline : 5th May

Use the pumping lemma to demonstrate that the following languages are not regular.

Q1. $L_1 = \{ w = 1^n : n \text{ is a power of two} \}$

Q2. $L_2 = \{ w = 1^n : n \text{ is a perfect cube (e.g., } n = 1, 8, 27 \dots) \}$

Q3. $L_3 = \{ ww^R \mid w \in \Sigma^* \text{ and } w^R \text{ means } w \text{ in reverse} \}$

Q4. $L_4 = \{ w_1 \# w_2 \text{ such that length of } w_1 = \text{length of } w_2 \}$

Q5. $L_5 = \{ w \in \Sigma^* \mid w = 0^i 1^j \text{ where } i < 3j \}$